WEIR - Assignment 1 - Preferential web crawling

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***Abstract*—In this assignment we developed a preferential web crawler for the news site 24ur.com that collects pages relevant to a user-specified topical query (for example, the conflict between Russia and Ukraine). Relevance is estimated by embedding page metadata, such as titles, summaries, and keywords, into a vector space using a pre-trained language model, then computing cosine similarity between the page embeddings and an embedding of the query. The crawler is comprised of an SQL database server, a RESTful API server that exposes database operations, and a client that uses browser automation to extract page content and hyperlinks and applies machine-learning scoring to prioritize crawl targets.**

I. Introduction

This report describes the design and implementation of a preferential web crawler developed to extract pages relevant to a user-specified topical query within a chosen domain. For this assignment, the chosen domain is the Slovenian news website 24ur.com.

The crawler comprises several components: a headless browser HTTP downloader and renderer, a data extractor for hyperlinks and images, a duplicate detector, a preferential URL frontier and a PostgreSQL database accessed via a RESTful API. The system is designed to run multiple parallel workers.

Pages are prioritised according to their relevance to a user-specified topical query (e.g. the conflict between Russia and Ukraine) by constructing a natural-language string from link metadata that is then converted into a vector representation using a pre-trained language model and ranked using cosine similarity.

The implementation also enforces canonical URLs, deduplication, respects robots.txt directives and crawl delay constraints and records pages, images and binary files found within the domain in the database.

The following sections document implementation details, parameter choices, performance evaluation, and visualisations of the resulting data.

II. Methodology

A. Crawler Architecture

The crawler is implemented as a modular system consisting of a RESTful API, a database server, and a client service. It is deployed using a Docker Compose project, with service configuration and crawling parameters defined in the Docker Compose project configuration file.

B. API Server and Database

The RESTful API service is implemented using a Python Flask[1] web server. It utilizes the SQLAlchemy[2] library to define and initialize the database schema in accordance with the schema provided in the assignment instructions.

The server supports configurable HTTP Basic authentication and uses Flask's built-in ad hoc SSL context to enable HTTPS, ensuring secure communication between components.

The crawler client interacts with the API using a Python module, which sends HTTP requests using the requests[3] library.

C. Crawler client

The crawler client uses the threading module to spawn a specified number of worker threads, which process URLs from a shared URL frontier. The Selenium [4] Firefox headless web browser is used to parse pages and extract metadata.

Links found on pages are extracted, scored, and added to the frontier. This process repeats until the desired number of pages has been crawled.

If an insertion into the database fails due to URL or content hash uniqueness constraints, the client handles the error by creating a page entry, marked as a duplicate.

The crawler respects the `robots.txt` file of the specified domain by adhering to the specified crawl delay.

D. Page parsing

Each subpage is parsed using the BeautifulSoup[5] module. The parser is used to extract article keywords and link URLs to other pages (in the `href` property or `onclick` JavaScript event handler).

The `onclick` event handler is processed by extracting URL candidates with regular-expression patterns that detect common JavaScript redirects as well as embedded JSON-like fields (`Url:`, `PlaybackUrl:`); if none match, it falls back to any quoted `.html` substring and returns the list of candidate URLs.

When saving a page to the database, each extracted URL is canonicalized using urllib[6] and tldextract[7]. Redirection links belonging to the domain `go-ctr-tracker.pub.24ur.si` are resolved using the `redir` key in the query string, the host is reduced to the highest level domain, the path is normalized, removing trailing slashes and redundant separators, common tracking parameters are removed, remaining query parameters are sorted deterministically, and the final URL is returned with the `https` scheme and without fragments.

Binary resources are detected by file extension (`.pdf`, `.doc`, `.docx`, `.ppt`, `.pptx`). For HTML pages, all outgoing `<a href>` targets are canonicalized and then checked for these extensions; matching targets are not added to the link frontier, but are instead stored as page data entries with their type code set to the corresponding type.

Images are detected in a similar way, using common image file extensions (`.jpg`, `.jpeg`, `.png`, `.gif`, `.webp`, `.bmp`, `.svg`), and are stored in the corresponding database table.

E. Preferential Crawling

Preferential crawling is implemented by maintaining a shared in-memory URL frontier as a priority queue. Each worker repeatedly removes the currently best-ranked URL and processes the corresponding page.

For each discovered URL, the crawler first applies canonicalization and filters out already visited URLs, off-domain URLs, and URLs disallowed by `robots.txt`. The parsing stage associates each link with metadata such as an inferred link title (from the URL slug), an optional summary (from nearby `p`/`span` elements with class `summary`), page-level article keywords (from the `article-keywords` block), and additional keyword-like text (`span` elements with the `label` class).

Since the same URL can be encountered multiple times from different pages and page regions, metadata is accumulated across occurrences in a per-URL list; every time new metadata is appended, a per-URL *version counter* is incremented.

Relevance scoring is performed with a multilingual sentence-transformer model (`paraphrase-multilingual-MiniLM-L12-v2` [8]). At the start of crawling the user query is embedded once, producing a normalized query embedding $\mathbf{e}_q$. For every candidate frontier URL u, the crawler constructs a short natural-language metadata string by concatenating de-duplicated fields in the form *Naslovi:* … *Povzetki:* … *Ključne besede:* … and embeds it into $\mathbf{e}_u$. The relevance score is then computed as cosine similarity

$$s(u) = \cos(\mathbf{e}_q, \mathbf{e}_u). \tag{1}$$

Candidates with no available metadata are assigned an empty metadata string and a neutral score of 0. Because the priority queue dequeues the smallest numeric key, the crawler converts similarity into a frontier priority by negation ($p(u) = -s(u)$), causing more relevant pages (larger s) to be scheduled earlier. To avoid using outdated scores, frontier entries carry the version number that was current when they were inserted; if a URL is encountered again and its metadata (and thus its version) increases, older queued entries are discarded when popped.

In addition to the in-memory frontier, every candidate is also stored in the database as a frontier page with its current priority, and a link edge is recorded from the currently crawled page to the corresponding frontier entry; when a frontier URL is later crawled, that database entry is updated with the full page content.

F. Deduplication

To avoid storing identical (or near-identical) content multiple times, every downloaded HTML page is assigned a content hash and the database enforces uniqueness of this hash. The hash is computed with the ppdeep[9] library.

Before hashing, the HTML is parsed and non-content elements (`script`, `style`, `noscript`) are removed. If the page contains an `<article>` element, its text is used as the primary content; two highly dynamic sub-blocks are explicitly removed when present: the comments placeholder (`id=comments_placeholder`) and a promotional widget (`data-upscore-zone=microsite_promo`). The remaining article text is extracted, whitespace-normalized, and hashed. For non-article pages, a fallback signature is constructed from stable navigation/text cues, such as submenu item titles and page headers (e.g., `h1` elements under `pageContent`).

The resulting hash is stored in the content hash field of the page table, which is protected by a unique constraint. When inserting or updating a page violates this constraint, the client resolves the conflict by locating the already-stored page with the same hash and creating a new entry marked as a duplicate (with the content cleared and the hash omitted to avoid retriggering the constraint). A link is then recorded from the duplicate entry to the original page, preserving the crawl graph while preventing redundant storage of page bodies.

III. Crawling Analysis

We conducted a performance analysis of our web crawler on a single target site. The crawler was configured with a maximum HTML-type page count of 5,100, four concurrent workers, and the query parameter: "Vojna med Ukrajino in Rusijo."

The results of the crawling process are summarized in Table I, highlighting the total number of pages retrieved, the count of duplicate pages, the total number of images extracted, and the average number of images per page.

TABLE I: Summary of crawler performance statistics

Total pages	Duplicates	Images	Avg. img./page	Avg. links/page
39,874	13	121,238	23.75	13.22

The crawler retrieved a total of 39,874 pages, of which only 13 were duplicates, indicating that it successfully avoided redundant content. The total number of images extracted was 121,238, resulting in an average of 23.75 images per page, which is understandable, as the target site is highly media-rich, with each page containing numerous visual elements. Additionally, the crawler discovered an average of 13.22 links per page, which indicates a well-connected site structure.

We also found that the page data table contains a single row, corresponding to a link to a PDF report, referenced in a news article. No other binary data was found.

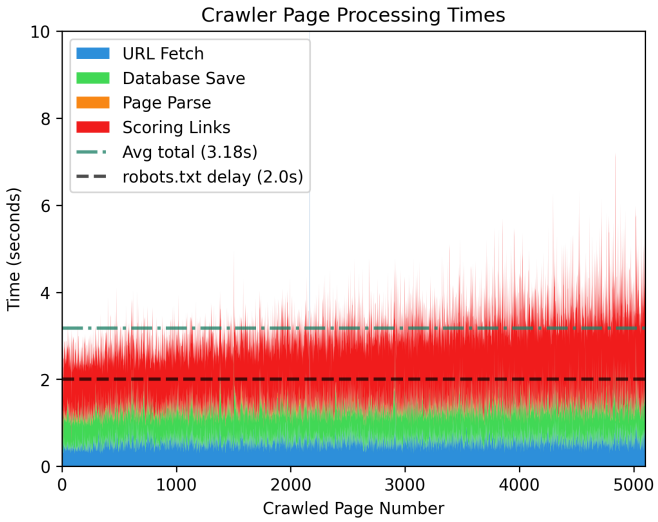

Fig. 1: Processing runtimes per page.

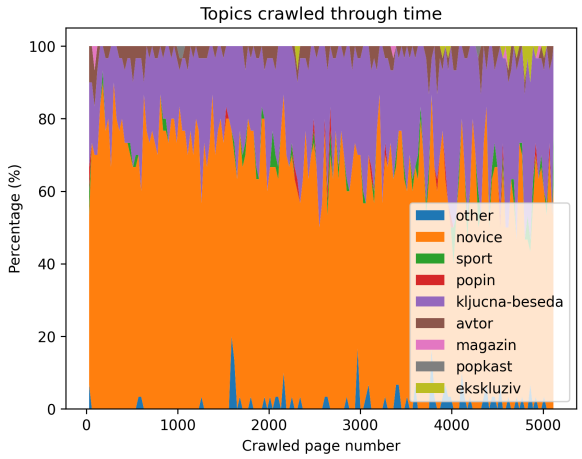

Fig. 2: Ratios of different topics as the number of crawled pages was increasing.

During crawling, we recorded processing times for individual pipeline stages, as shown in Figure 1. The results indicate that overall crawling time is dominated by the BERT-based scoring of URL metadata, suggesting that optimizing this component would yield the most significant performance improvements. Additionally, we observe a gradual increase in scoring time as the number of processed pages grows. This trend is likely due to a combination of hardware effects, such as thermal throttling, and the increasing size of internal data structures used to manage the crawling process.

The green line in Figure 1 represents the crawl delay specified in the *robots.txt* file of the target site. This value defines the minimum allowable time between consecutive page fetches, effectively imposing a lower bound on the crawler's runtime. During the first half of the 5100-page crawl, the system operates close to this theoretical limit, indicating that the pipeline is able to sustain near-optimal throughput under these constraints.

A. Crawling Quality

To assess the quality of the crawler, we performed a semi-manual evaluation by constructing an interactive data map in which HTML-type pages are represented as two-dimensional points. This visualization enables inspection of the semantic structure of the collected data and helps identify potential inconsistencies or noise in the crawling process.

The pipeline begins by encoding page content using TF-IDF with a vocabulary size of 15,000. To reduce the impact of high-frequency noise and compress the representation, the vectors are projected into a 40-dimensional space using truncated SVD. Clustering is then performed on these reduced representations using the K-means algorithm, where the optimal number of clusters is selected based on the silhouette score.

For visualization, the clustered data is embedded into two dimensions using UMAP, which preserves local and global structure sufficiently for qualitative inspection. To improve interpretability, each cluster is labeled with representative keywords. This is achieved by concatenating the text of all documents within a cluster into a single corpus, applying TF-IDF to extract 30 candidate keywords, and then re-ranking them using a BERT-based semantic similarity model. The top six keywords are selected to describe each cluster.

The Figure 4 presents the complete data map, where each point corresponds to an individual article that can be interactively opened in a browser. Points are annotated with cluster identifiers, and the associated keywords for each cluster are listed alongside the visualization to aid interpretation. Using the silhouette score as a selection criterion, the data was partitioned into 40 clusters.

The Figure 5 highlights a subset of clusters that were identified as relevant to the query. This selection was performed by manually inspecting the cluster keywords and determining their semantic alignment with the query topic. In total, 23 clusters were marked as relevant, collectively accounting for 55.74% of all data points.

B. 24Ur.com Topography

We conducted a focused categorical and topographical analysis of our crawl of *24Ur.com*. Specifically, we examined a sequence of 5,100 consecutively crawled pages and analyzed the distribution of link topics within them. As shown in Figure 2, the crawl is strongly dominated by links belonging to the *novice* category, while links under *kljucna-beseda* constitute the second largest proportion. This distribution is consistent with the query parameter, as articles related to international conflicts are predominantly categorized within these topics. As the number of crawled pages increases, the *novice* category slightly shrinks as the crawler has to start exploring other unvisited alternatives (*ekskluziv*, more of *kljucna-beseda* and *other*).

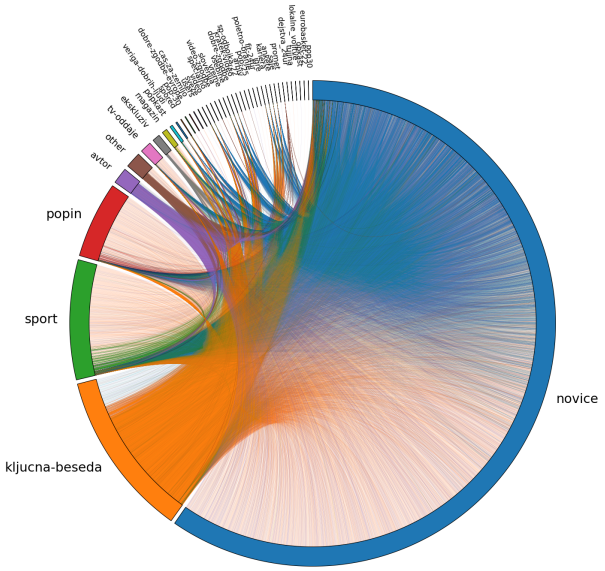

Fig. 3: Category links.

To visualize all 5,100 crawled pages and their links, we plotted a circular link diagram 3. Pages were groupped based on the part of the URL after the domain - topics. Links were drawn in the color of the topic that the source article is on, showing how often each topic referenced others.

Most pages are articles from the *novice* topic, which is what we hoped for. Each article contains some keywords, which have their own pages, listing other articles with the same keywords. For our test query, *šport* articles are second most common as Russia is also very involved in sports.

Avtor pages are a useful source of relevant articles, because each author usuall writes articles on specific topics. That's why *avtor* pages have lots of outgoing links. However, we could only extracted the first 60 articles from each author since that's the maximum number of displayed articles on any *avtor* page.

IV. Disscussion

The crawler achieved a large, diverse crawl of the target domain while respecting domain constraints and prioritizing topical relevance. The low duplicate count suggests that the content-hash based deduplication is effective for the target site, while the very small number of binary resources indicates that most relevant material is delivered as HTML. The preferential strategy also appears to improve topical focus: the clustering analysis shows that more than half of the collected pages fall into query-aligned semantic clusters, which is consistent with the intended behavior of the scoring and frontier prioritization.

There are, however, limitations and opportunities for improvement. The use of a headless browser increases robustness on dynamic pages but is computationally expensive and limits crawl throughput. The relevance model is based on short metadata strings; if metadata is missing or noisy, scores may be less reliable. URL canonicalization was satisfactory, but had unusual edge cases with query strings.

Database error handling could have been done on the server side instead of client for faster communication. Some database queries weren't equipped with rollback options in case of database insertion errors. Sending frontier pages to the database was at first implemented roughly with batches, but later changed to sending individual ones due to risks of concurrent writing and difficulty of error handling.

Future work should focus on improving both the efficiency and robustness of the crawling pipeline. In particular, the server component could be enhanced through server-side conflict resolution and reintroducing batch-based frontier insertion with proper concurrency control, enabling more efficient communication and reduced overhead. URL canonicalization should be extended to handle query parameters more consistently. Additionally, the metadata extraction and scoring pipeline could be improved by generating more context-rich yet compact representations.

V. Conclusion

We implemented a preferential web crawler for a target website using a modular architecture with a RESTful API, PostgreSQL storage, and a multi-threaded Selenium-based client. The crawler respects robots.txt, canonicalizes URLs, performs deduplication, and prioritizes pages by semantic relevance to a user query using a language model. Our analysis of a 5,100 page crawl indicated low duplicate rates and strong topical concentration, as confirmed by clustering-based quality analysis.

Overall, the system meets the assignment requirements and demonstrates how relevance-driven frontier scheduling can improve the usefulness of a focused crawl.

References

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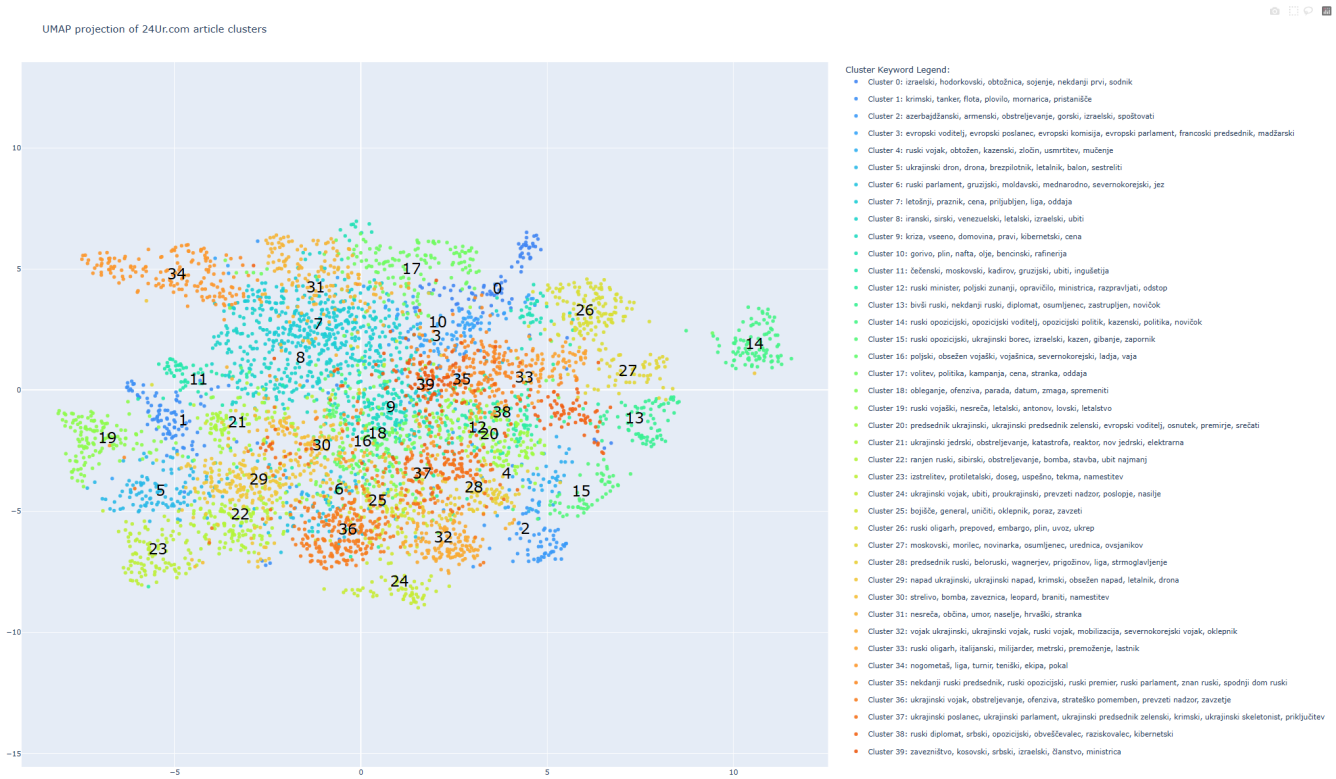

Fig. 4: UMAP projection of all crawled HTML pages grouped into semantic clusters. Each point represents an article, with cluster identifiers and associated keywords shown for interpretation.

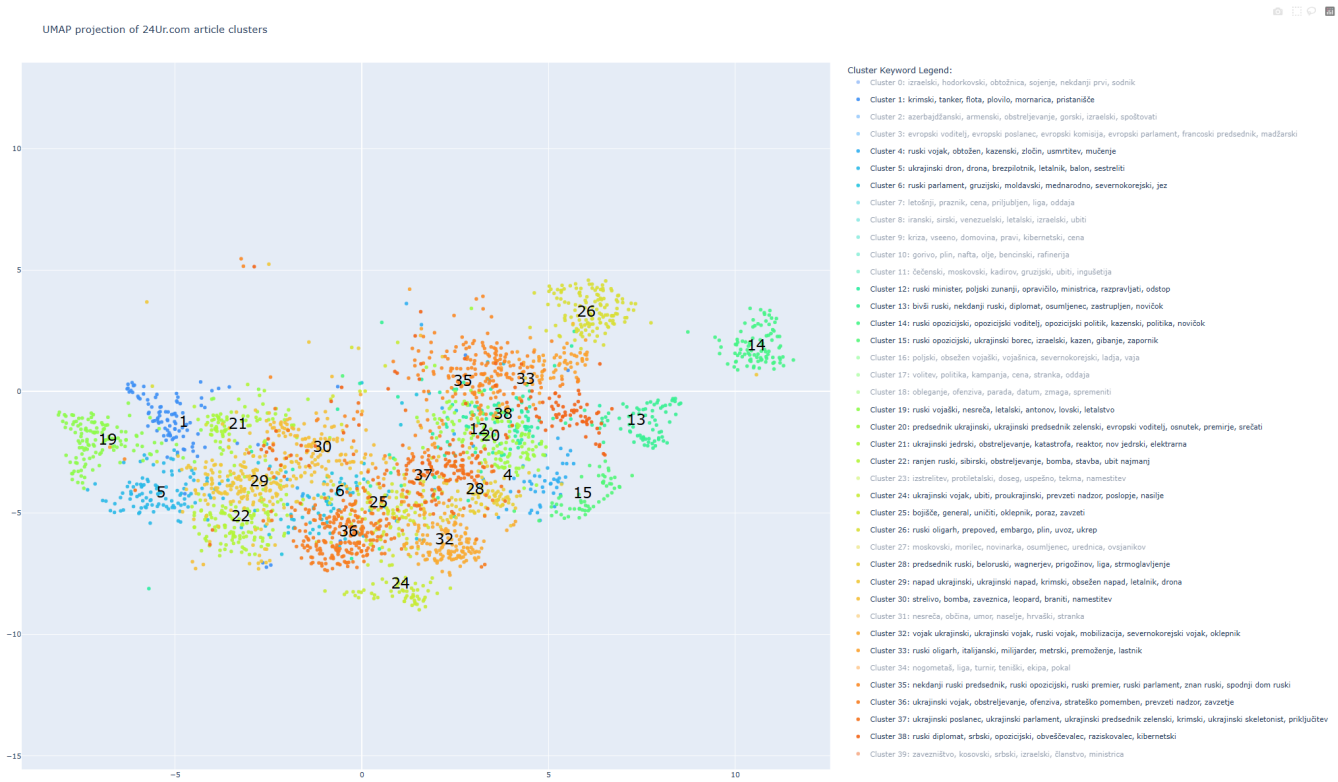

Fig. 5: UMAP projection highlighting clusters relevant to the query. Selected clusters were manually identified based on semantic alignment of their keywords with the query.